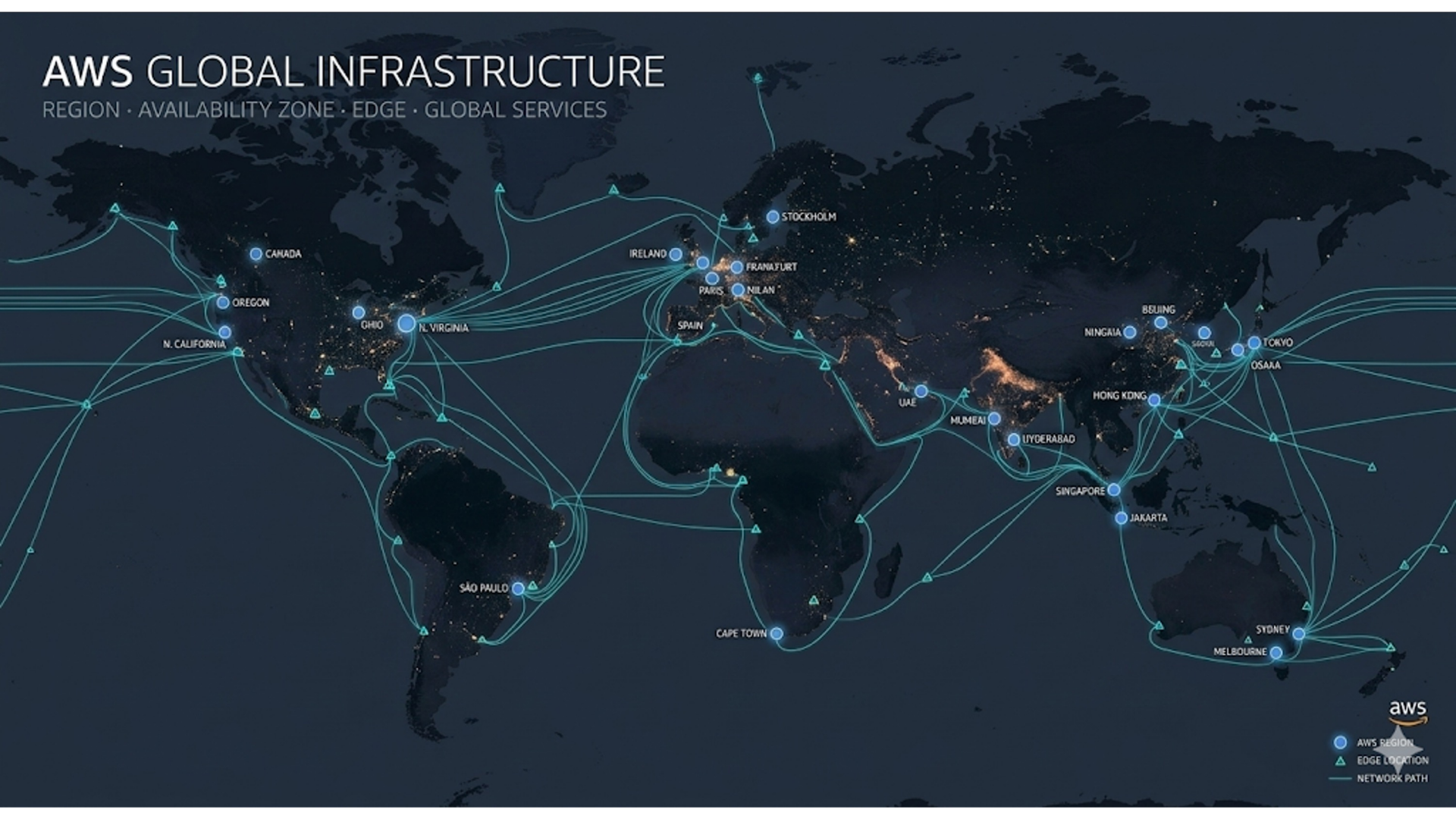
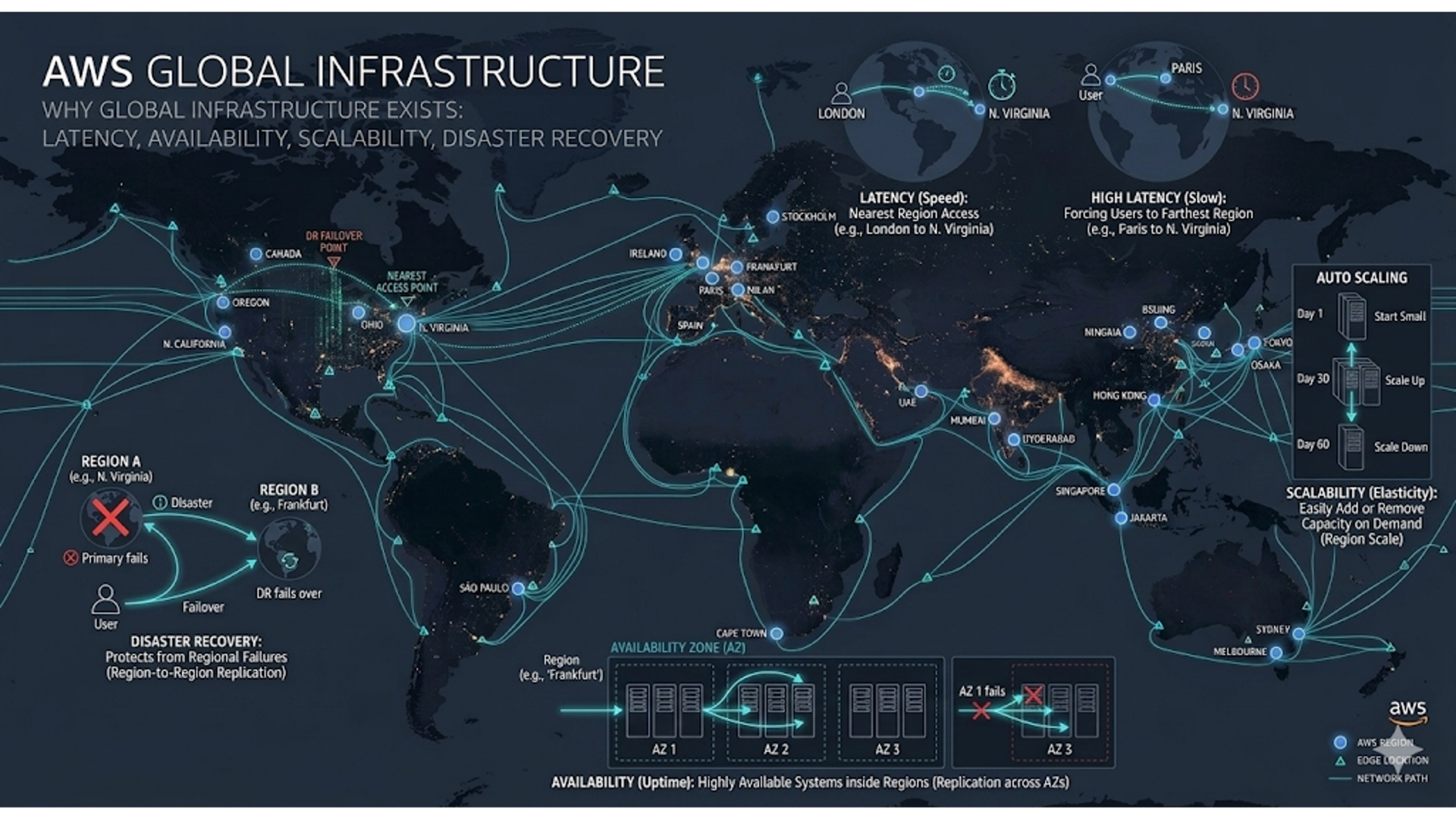

AWS GLOBAL INFRASTRUCTURE
REGION · AVAILABILITY ZONE · EDGE · GLOBAL SERVICES
STOCKHOLM
CANADA
IRELAND
FRANKFURT
PARIS
MILAN
OREGON
OHIO
N. VIRGINIA
N. CALIFORNIA
SPAIN
BEIJING
NINGXIA
SEOUL
TOKYO
OSAKA
UAE
HONG KONG
MUMBAI
HYDERABAD
SINGAPORE
JAKARTA
SÃO PAULO
CAPE TOWN
SYDNEY
MELBOURNE
aws
AWS REGION
EDGE LOCATION
NETWORK PATH

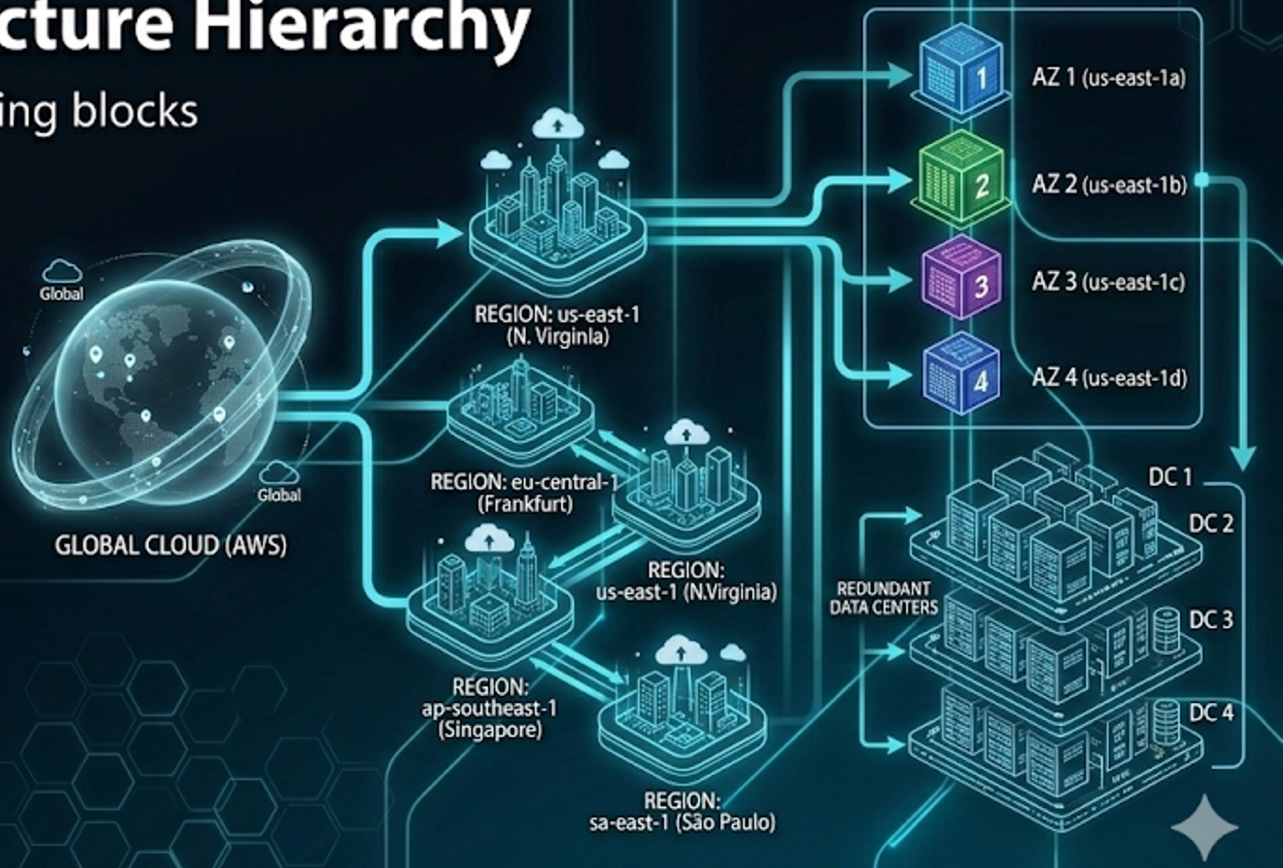

AWS GLOBAL INFRASTRUCTURE
WHY GLOBAL INFRASTRUCTURE EXISTS:
LATENCY, AVAILABILITY, SCALABILITY, DISASTER RECOVERY
LONDON
N. VIRGINIA
User
PARIS
N. VIRGINIA
LATENCY (Speed):
Nearest Region Access
(e.g., London to N. Virginia)
HIGH LATENCY (Slow):
Forcing Users to Farthest Region
(e.g., Paris to N. Virginia)
DR FAILOVER POINT
NEAREST ACCESS POINT
CANADA
OREGON
OHIO
N. VIRGINIA
N. CALIFORNIA
STOCKHOLM
IRELAND
FRANKFURT
PARIS
MILAN
SPAIN
UAE
MUMBAI
HYDERABAD
BEIJING
NINGXIA
SEOUL
TOKYO
OSAKA
HONG KONG
SINGAPORE
JAKARTA
SÃO PAULO
CAPE TOWN
SYDNEY
MELBOURNE
AUTO SCALING
Day 1
Start Small
Day 30
Scale Up
Day 60
Scale Down
SCALABILITY (Elasticity):
Easily Add or Remove
Capacity on Demand
(Region Scale)
REGION A
(e.g., N. Virginia)
Disaster
Primary fails
REGION B
(e.g., Frankfurt)
DR fails over
Failover
User
DISASTER RECOVERY:
Protects from Regional Failures
(Region-to-Region Replication)
AVAILABILITY ZONE (AZ)
Region
(e.g., 'Frankfurt')
AZ 1
AZ 2
AZ 3
AZ 1 fails
AZ 3
AVAILABILITY (Uptime): Highly Available Systems inside Regions (Replication across AZs)
aws
AWS REGION
EDGE LOCATION
NETWORK PATH

# AWS Infrastructure Hierarchy

Understanding the building blocks

**Hierarchy:**

- Global Cloud
- Regions
- Availability Zones
- Data Centers



# GLOBAL VS REGIONAL SERVICES

Understanding the Service Scope

## GLOBAL SERVICES

- IAM (Identity & Access Management)
- Route53 (DNS Service)
- CloudFront (CDN & Edge Caching)

## REGIONAL SERVICES

- EC2 (Virtual Servers)
- RDS (Relational Database)
- S3 (Object Storage)

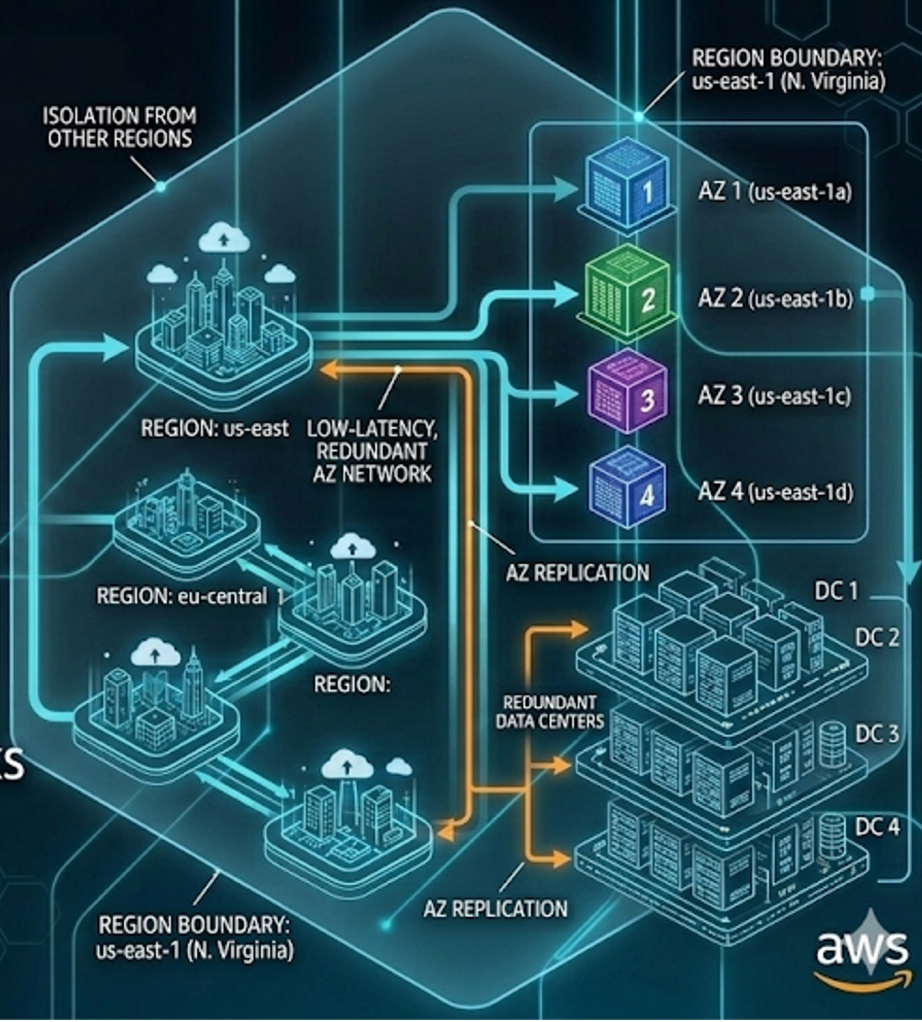
# WHAT IS A REGION?

**REGION** = Geographic area with multiple AZ

## Region Properties:

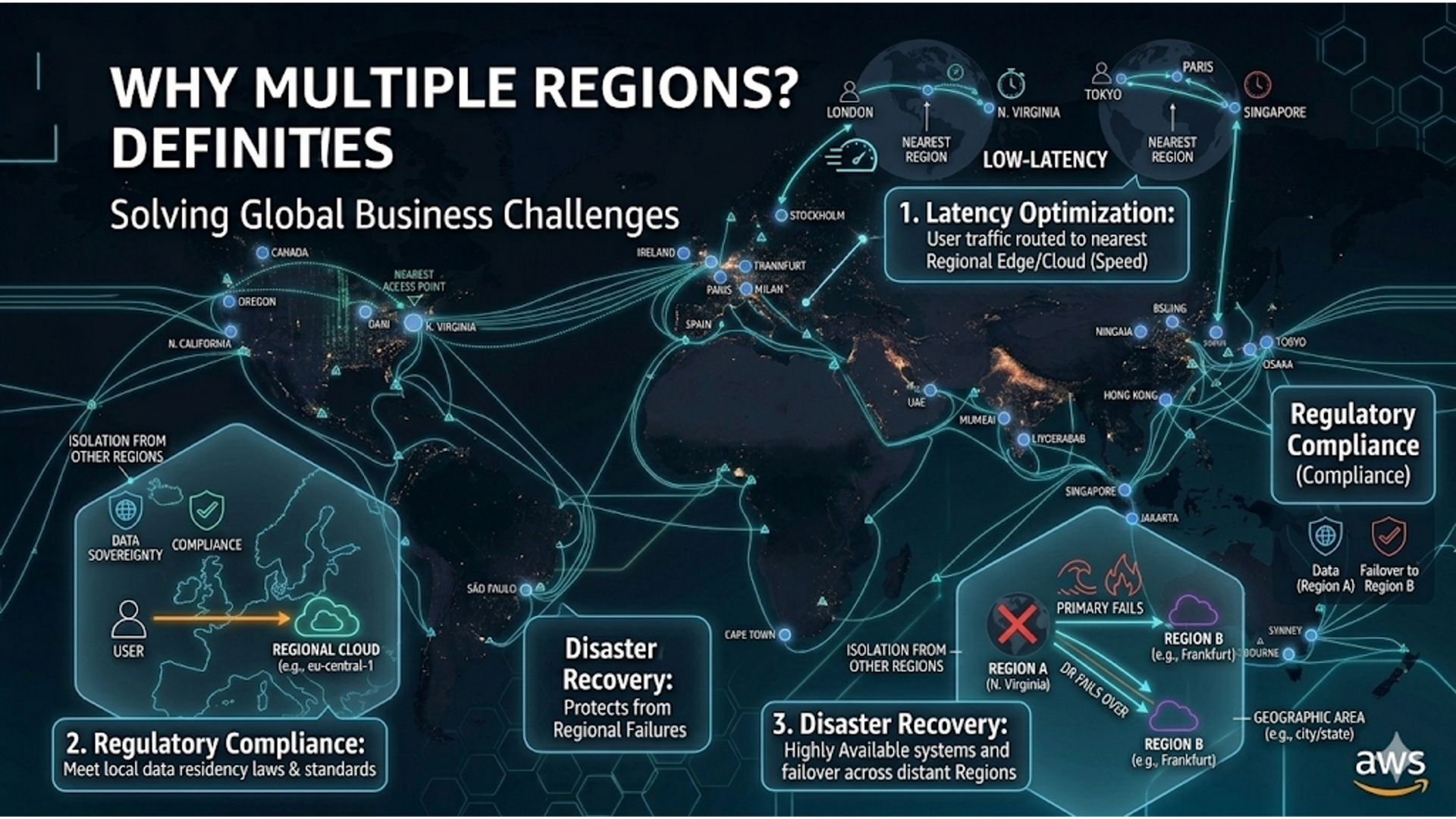
- Geographic boundary
- Fully isolated from other Regions
- Multiple Availability Zones (AZ)
- Connected by low-latency networks

EXAMPLE: “us-east-1” (N. Virginia)

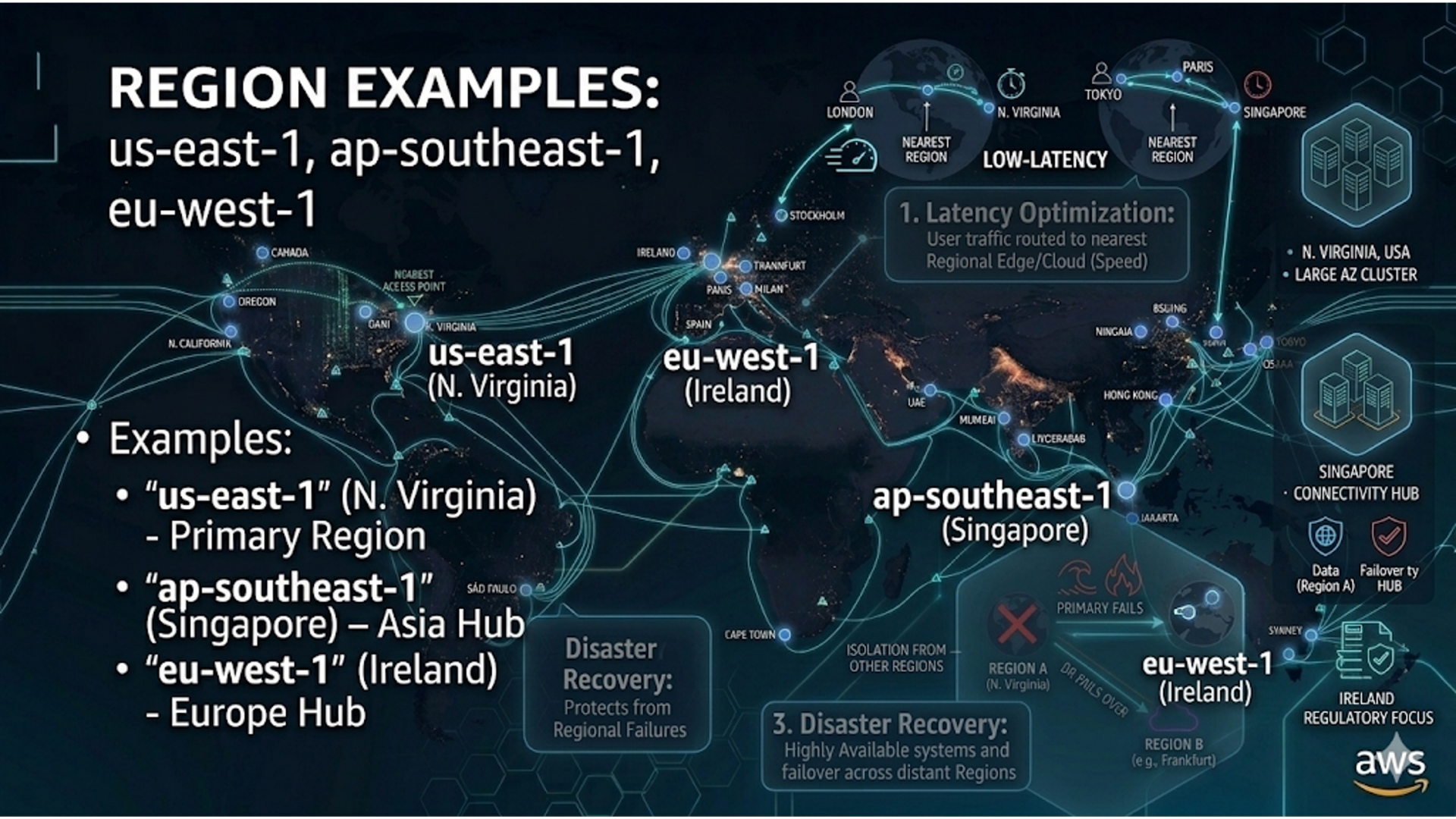


# WHY MULTIPLE REGIONS? DEFINITIES

## Solving Global Business Challenges



# REGION EXAMPLES:
us-east-1, ap-southeast-1, eu-west-1

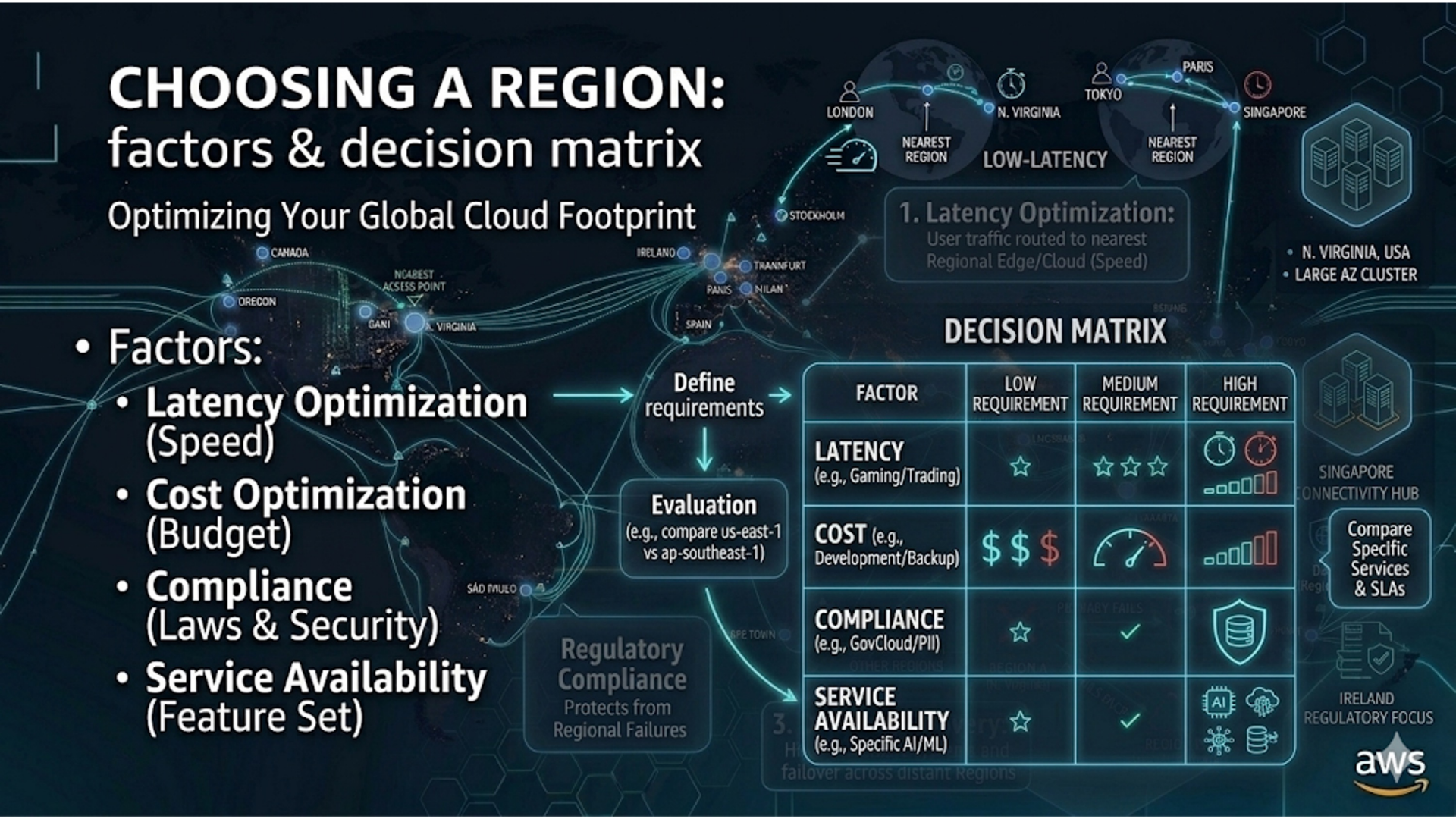



- Examples:
  - "**us-east-1**" (N. Virginia) - Primary Region
  - "**ap-southeast-1**" (Singapore) – Asia Hub
  - "**eu-west-1**" (Ireland) - Europe Hub

# CHOOSING A REGION:
## factors & decision matrix

Optimizing Your Global Cloud Footprint

- Factors:
  - **Latency Optimization** (Speed)
  - **Cost Optimization** (Budget)
  - **Compliance** (Laws & Security)
  - **Service Availability** (Feature Set)




| FACTOR | LOW REQUIREMENT | MEDIUM REQUIREMENT | HIGH REQUIREMENT |
|---|---|---|---|
| **LATENCY** (e.g., Gaming/Trading) | ☆ | ☆ ☆ ☆ | |
| **COST** (e.g., Development/Backup) | $ $ $ | | |
| **COMPLIANCE** (e.g., GovCloud/PII) | ☆ | ✓ | |
| **SERVICE AVAILABILITY** (e.g., Specific AI/ML) | ☆ | ✓ | AI |

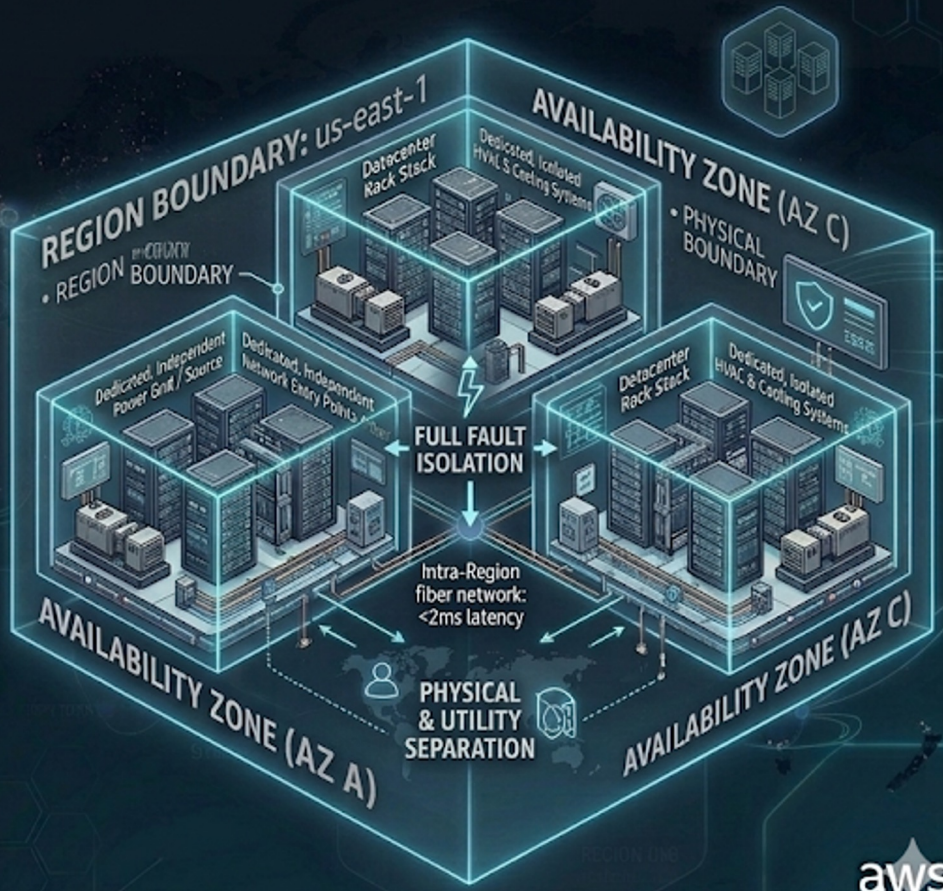
# WHAT IS AN AVAILABILITY ZONE (AZ)? ZONE (AZ)?

- AZ = isolated datacenter cluster within region.
- **Key takeaways**
  1. Full Datacenter Fault Isolation
  2. Multiple AZs per Region
  3. Redundant, Low-Latency Internal Network

**Final thought:** AZs are the foundation for building highly resilient, scalable systems.

# AZ ISOLATION

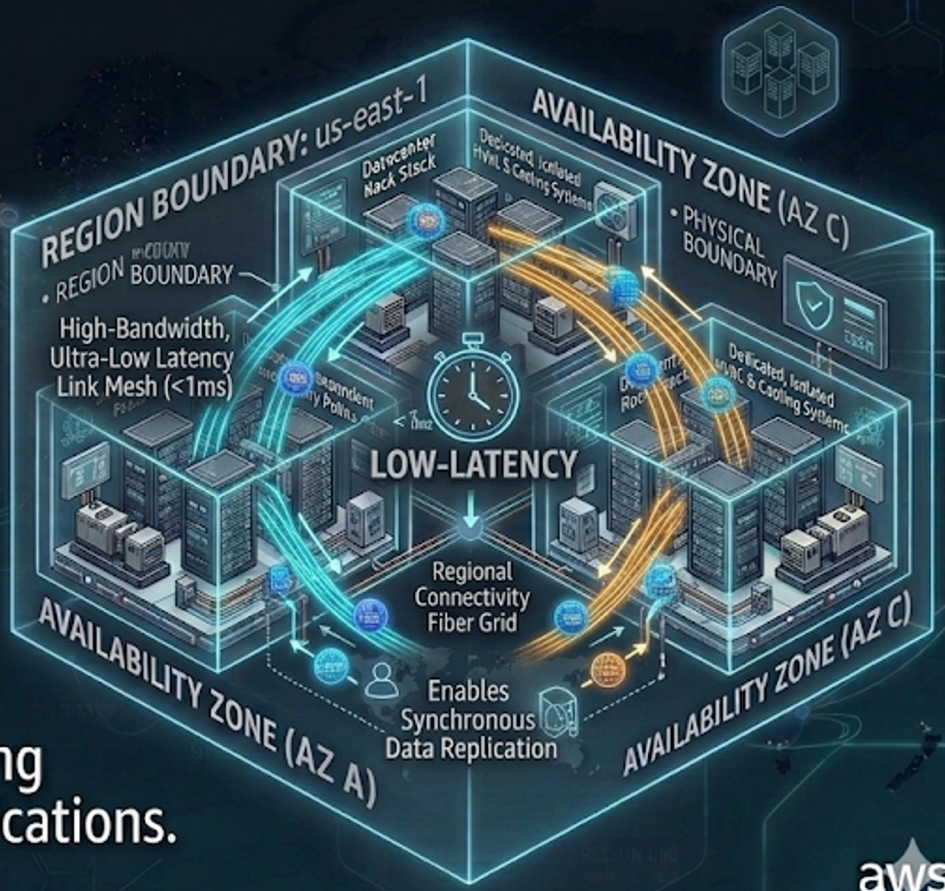
- AZs have:
  - separate power
  - separate networking
  - separate cooling




**Final thought:** AZs are the foundation for building highly resilient, scalable systems.

# AZ Connectivity

- High bandwidth connections between AZs.
- Key takeaways
  1. Ultra-Low Latency (<1ms)
  2. Redundant, High-Bandwidth Intra-Region Network
  3. Enables Synchronous Data Replication

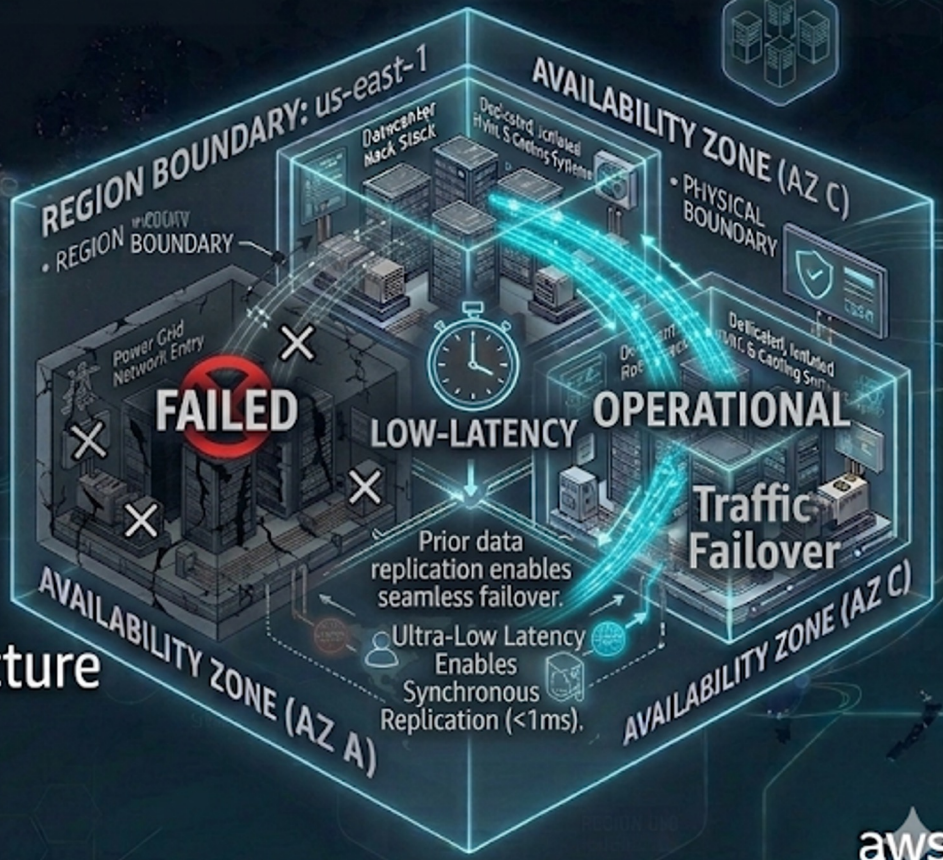



**Final thought:** High-speed AZ connectivity is essential for building highly available, low-latency applications.

# Why Multi-AZ Matters

## Scenario: Single AZ Failure

- A physical or utility failure isolates one AZ (e.g., AZ A).
- High availability mechanisms detect the failure.
- Application traffic automatically fails over to the healthy AZ (e.g., AZ C).




**Final thought:** Multi-AZ architecture ensures resilience, preventing downtime for your applications.

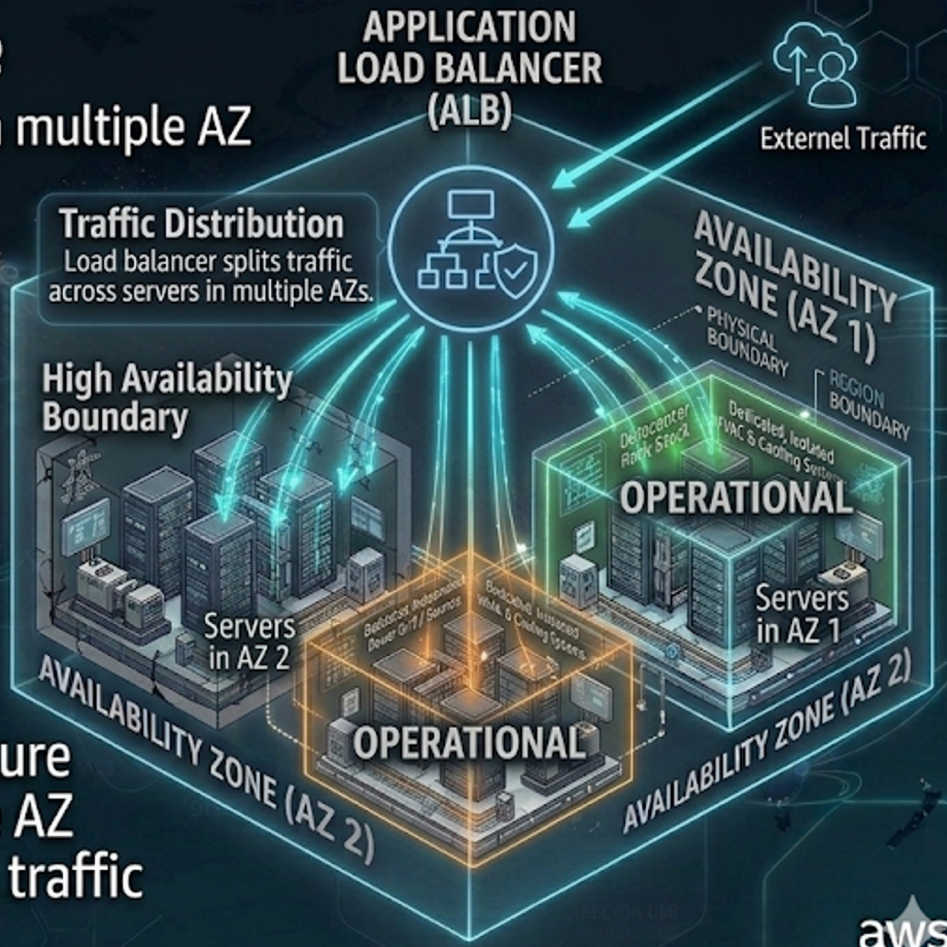
# Multi-AZ Architecture

Example: Load balancer → instances in multiple AZ

## Standard High Availability Setup

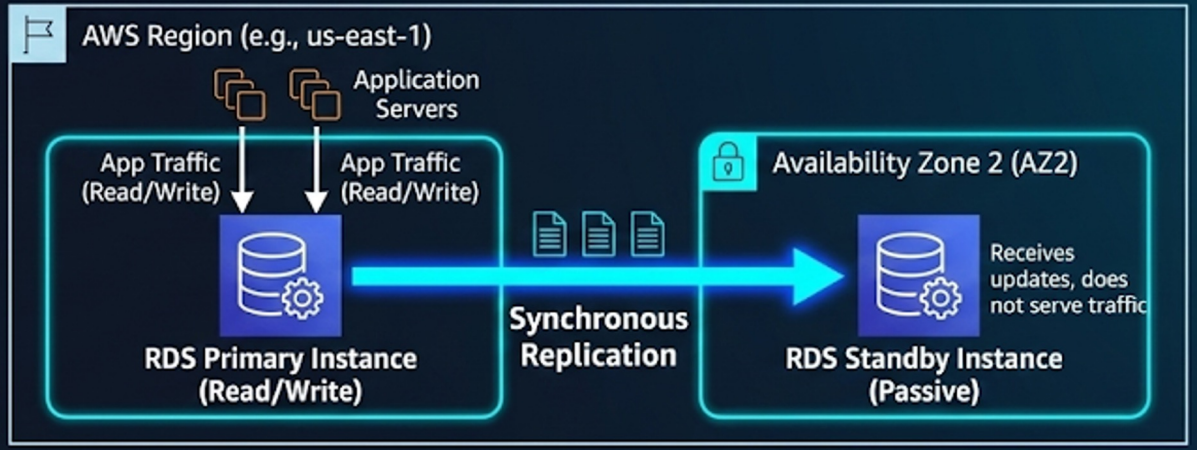
- Application Load Balancer (ALB) acts as the single entry point.
- Redundant servers/instances are running in separate, isolated Availability Zones (AZs).
- ALB continuously distributes traffic, spreading load and ensuring service continuity.

**Final thought:** Multi-AZ architecture prevents downtime during a single AZ failure by automatically re-routing traffic to healthy servers in another zone.

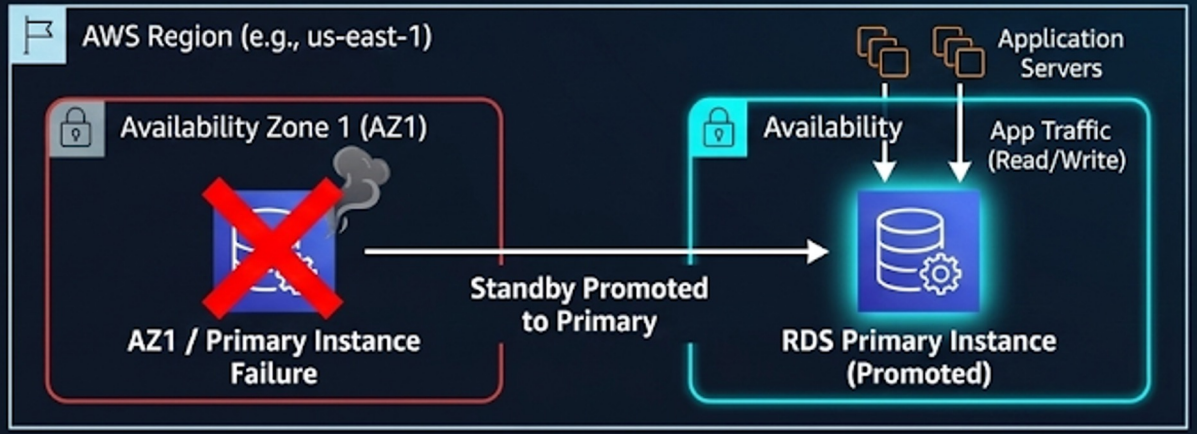


# Amazon RDS Multi-AZ Deployment

**Normal Operation**
(High Availability)




**Automatic Failover Scenario**




Automatic Failover detects failure and promotes standby, ensuring minimal downtime.

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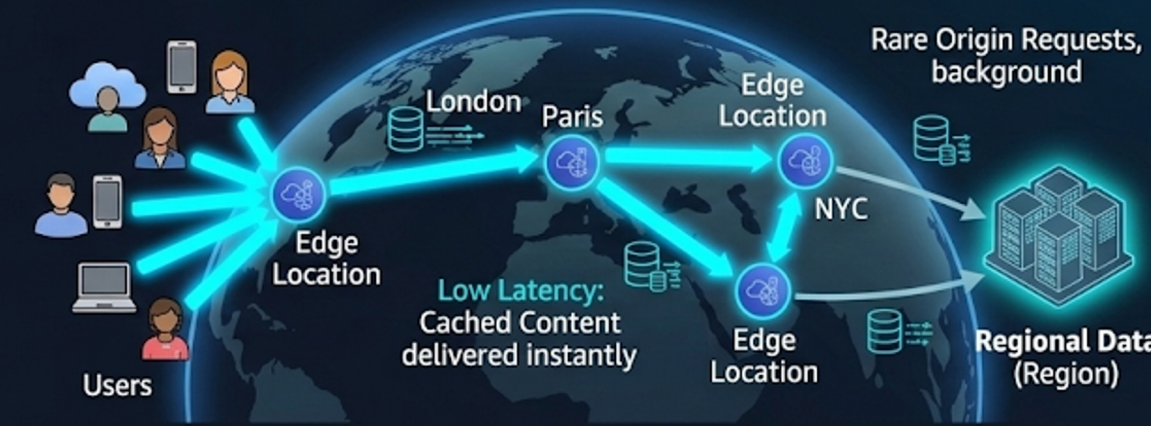
# WHAT IS AN EDGE LOCATION?

**Edge** = CDN location near users.



# CLOUDFRONT CONCEPT FLOW

USERS / CLIENTS

1. Request

2.

LOW-LATENCY:
CACHED CONTENT
DELIVERED INSTANTLY
CONTENT TO USER

EDGE LOCATION
(CDN NODE)

CACHE

YES

CACHE HIT

NO

CACHE MISS

5. Edge Location updates Cache

Cache Miss Path
(Higher Latency)

3.

4. Origin
Fetches
Content

ORIGIN
(e.g., S3, EC2)

Cache Hit Path
(Low Latency)

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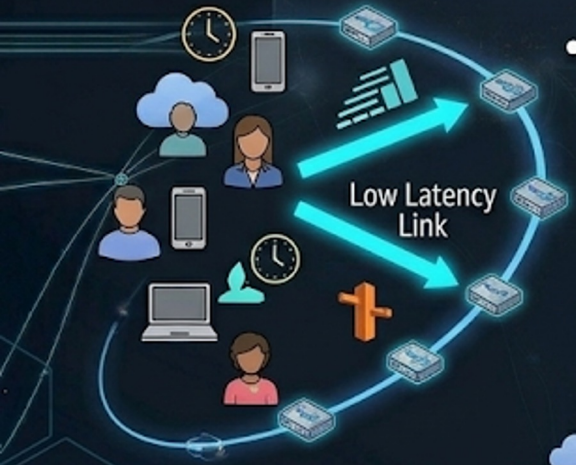
# EDGE vs REGION: Comparison

CDN

## EDGE LOCATION

speed

low latency

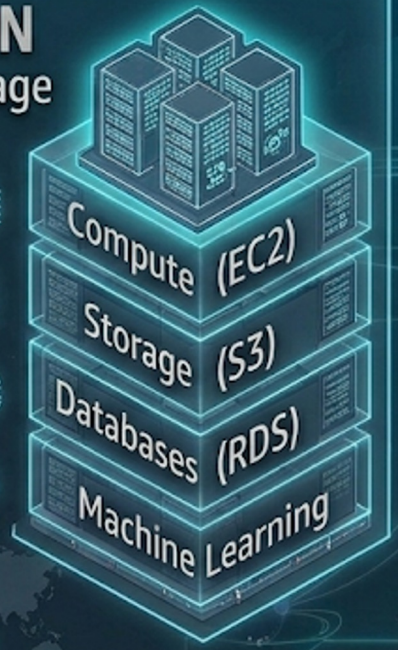



- Global network for speed.
- Route53 (DNS)
- CloudFront (CDN)
- Performance focus (Latency)

## AWS REGION

compute + storage

- Full suite of AWS services.
- Multiple Availity Zones (AZ).
- Capability focus (Services)




Low Latency

**Latency** vs **Capability**

Full Services

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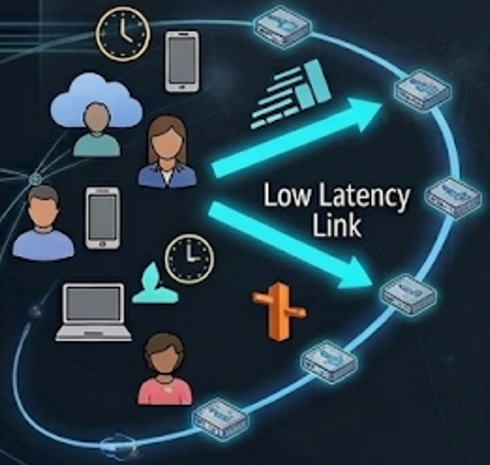
# EDGE USE CASES: Infographic Slide

Speed, Availability, and Security

## CONTENT DELIVERY NETWORK (CDN)

**CDN: CloudFront** (Cached Content)




- Faster load times via globally distributed nodes.
- Reduced origin server load & bandwidth costs.
- Reduced origin server load & bandwidth costs.
- Deliver diverse static & dynamic content (images, video, data).

## GLOBAL DNS

**DNS: Route53** (Global Resolvers)

- Faster recursive DNS resolution via closest resolver.
- High availability with globally distributed recursive resolvers.
- Support advanced routing policies (latency, geolocation).

## DDoS PROTECTION: (Edge Mitigation)

- Distribute and absorb large-scale malicious traffic volumes globally. A nuits a powerful defense barrier
- Analyze and filter malicious traffic closest to the source.
- Ensure service availability during attack events.

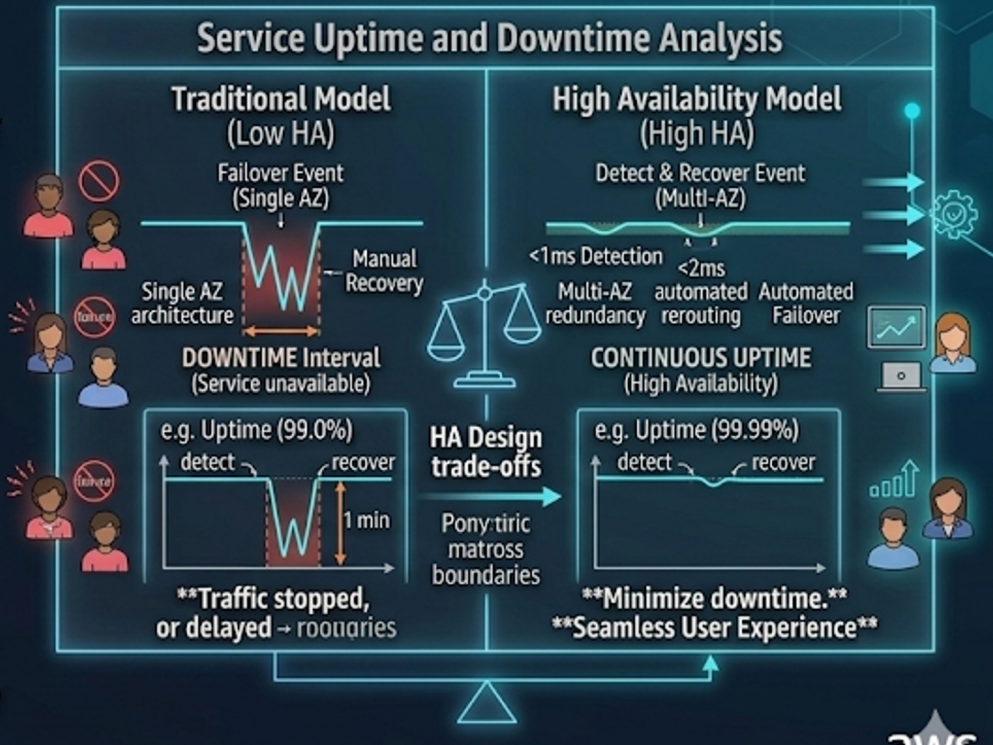
**Final thought:** Leveraging Edge Use Cases optimizes performance, increases availability, and enhances the security posture for globally distributed applications.

# WHAT IS HIGH AVAILABILITY

## Understanding Reliability and Resiliency

- Goal: Minimize downtime.
- Redundant systems automatically detect and recover from failures.
- Application traffic is seamlessly rerouted during failover.
- Designed to provide maximum service uptime and accessibility (e.g., 99.99%).

**Final thought:** High availability architecture is critical for business continuity and providing a reliable service to your customers.

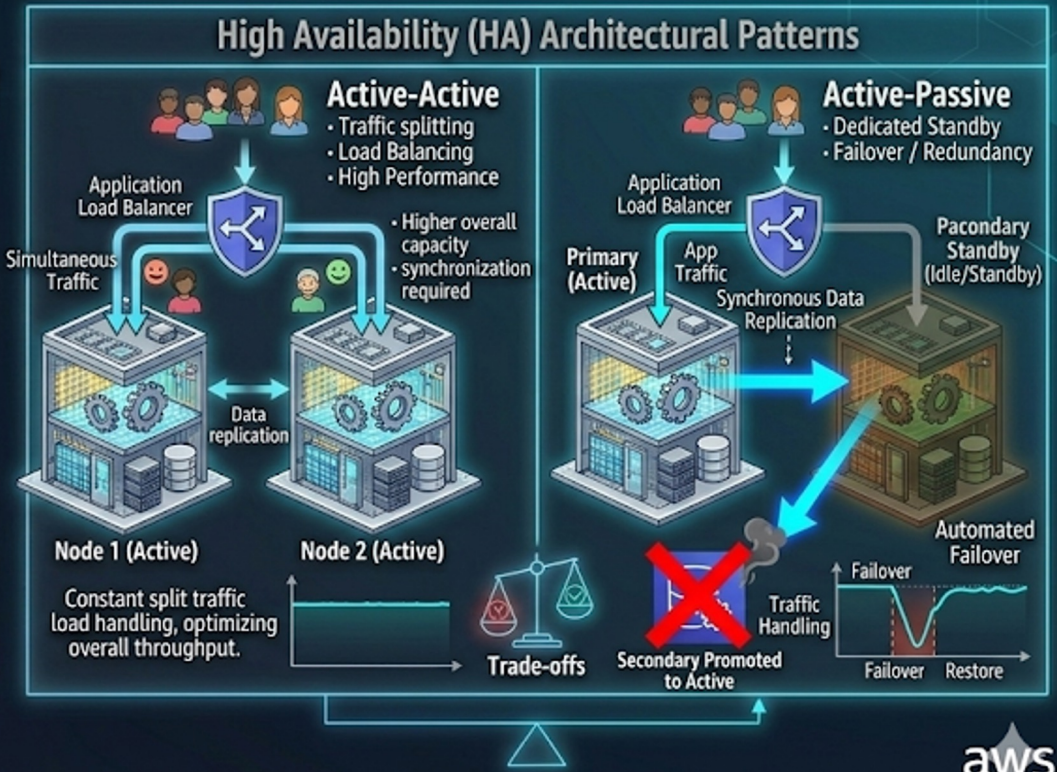


# HA Patterns

## Common architectural strategies

- Redundancy is the foundation for high availability.
- Traffic management and data sync are critical considerations.
- Patterns vary in resource utilization and failure response.

**Final thought:** The chosen HA pattern impacts scalability, performance, complexity, and cost for system architecture.



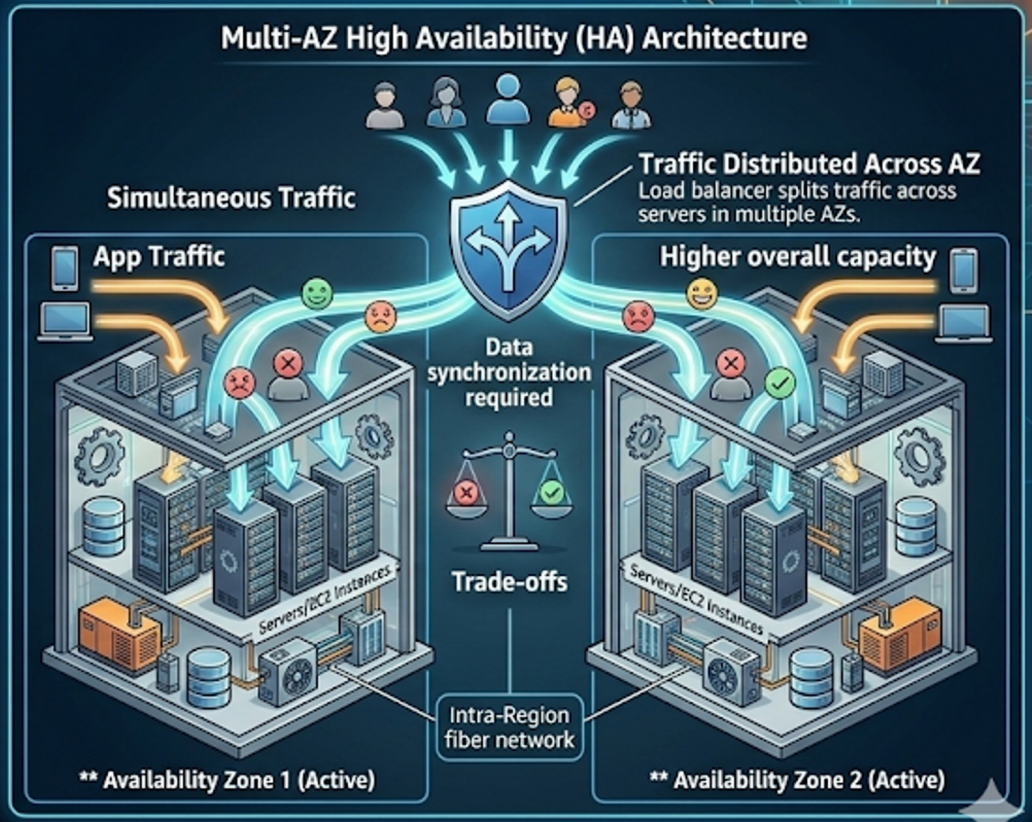
# Multi-AZ HA

## Traffic distributed across AZ

### seamless Failover & Resilience

- Multi-AZ redundancy prevents downtime during a single AZ failure.
- High availability (HA) mechanisms automatically detect failures.
- Application Load Balancers (ALB) seamlessly re-route traffic to healthy zones.
- Architecture designed to prevent data center failure impacting service availability.

**Final thought:** Multi-AZ HA architecture is essential for building mission-critical, resilient, and reliable systems.



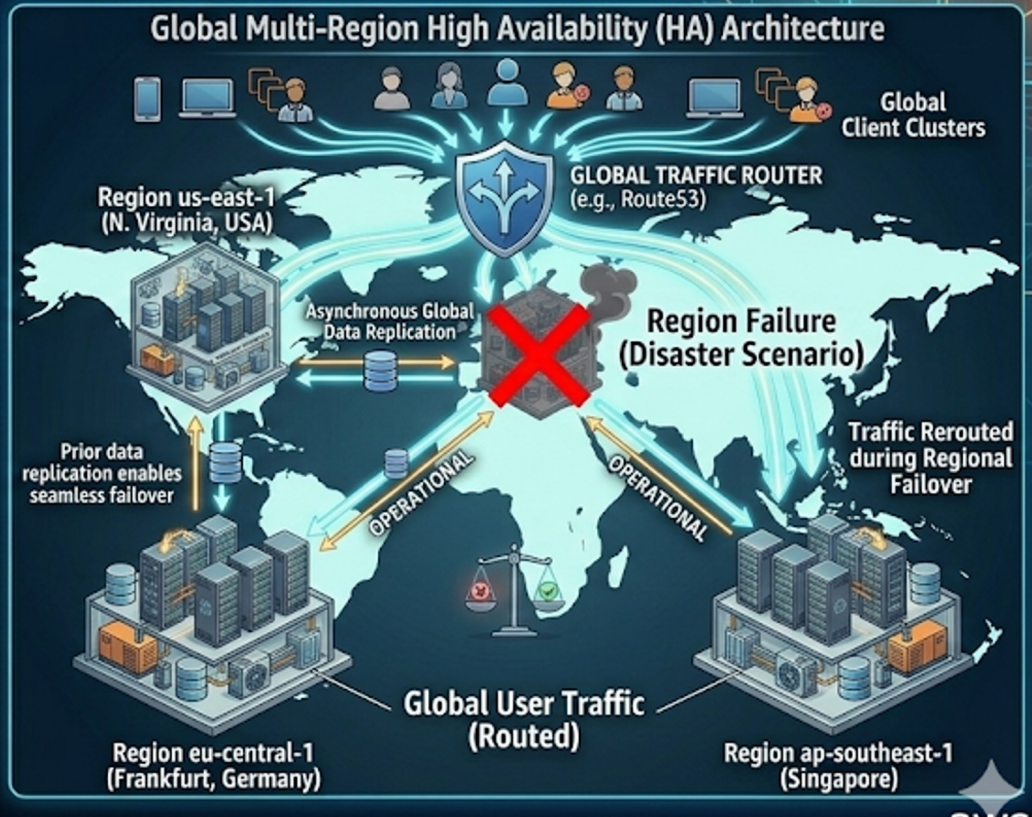
# GLOBAL RESILIENCE: Multi-Region HA

## Traffic Routed Globally with Seamless Failover

### Global Failover & Seamless Operations

- Multi-Region redundancy prevents global downtime during a major regional failure.
- Global Traffic Management automatically detects region failure and re-routes users to healthy regions.
- Data replication across global regions ensures minimal data loss and rapid recov
- Design isolates regions to prevent cascading failures across the global network.

**Final thought:** Multi-Region HA architecture is critical for global disaster recovery and ensuring minimal impact from catastrophic events.



# DISASTER RECOVERY STRATEGY LADDER

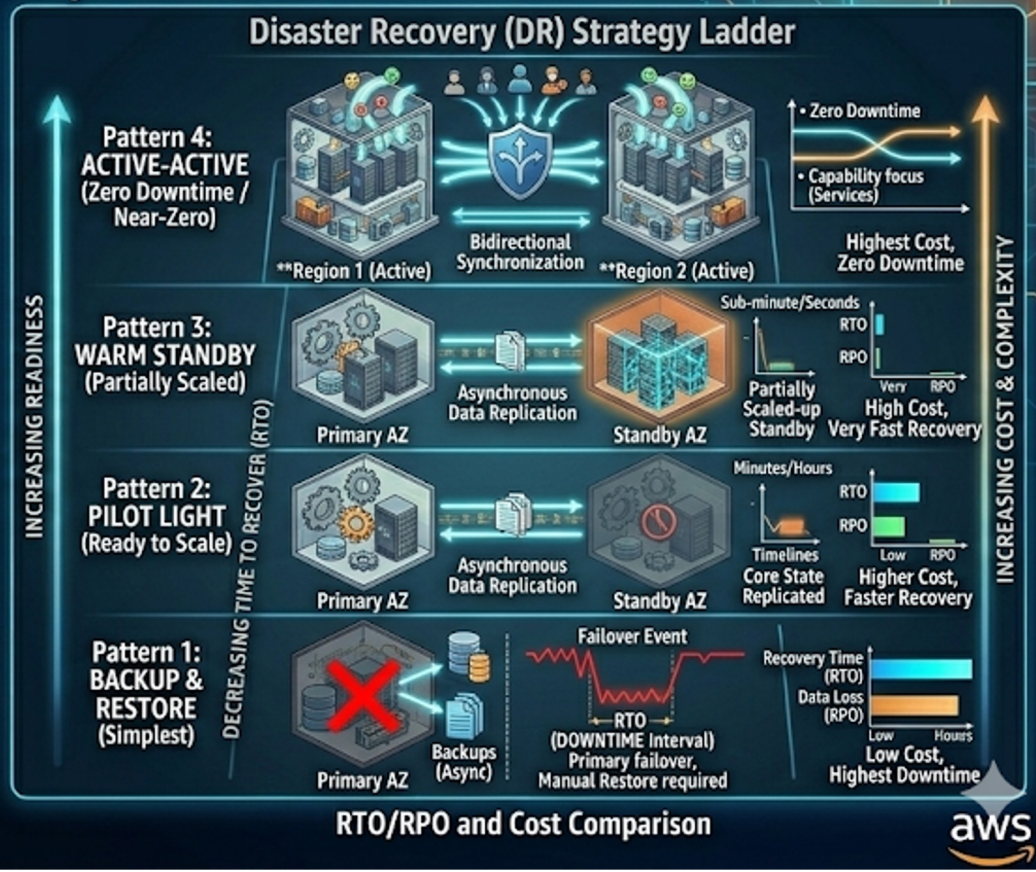
## Evaluating Readiness, Cost, and Recovery

## DR stategies

### Evaluating Recovery Time Objective (RTO) & Recovery Point Objective (RPO)

- **Failure scenario:** catastrophic regional event isolates multiple AZ or regions.
- **DR objective:** protect from catastrophic regional failures.
- Strategies have distinct trade-offs in in recovery speed and complexity.
- Aim is to restore operations according to defined RTO/RPO.

**Final thought:** Multi-region architecture ensures reghor, recovery, mhan-cnlision, critical, and top raint foregement.



# MULti-AZ: Analysis and Trade-offs

## Understanding High Availability Within a Single Region

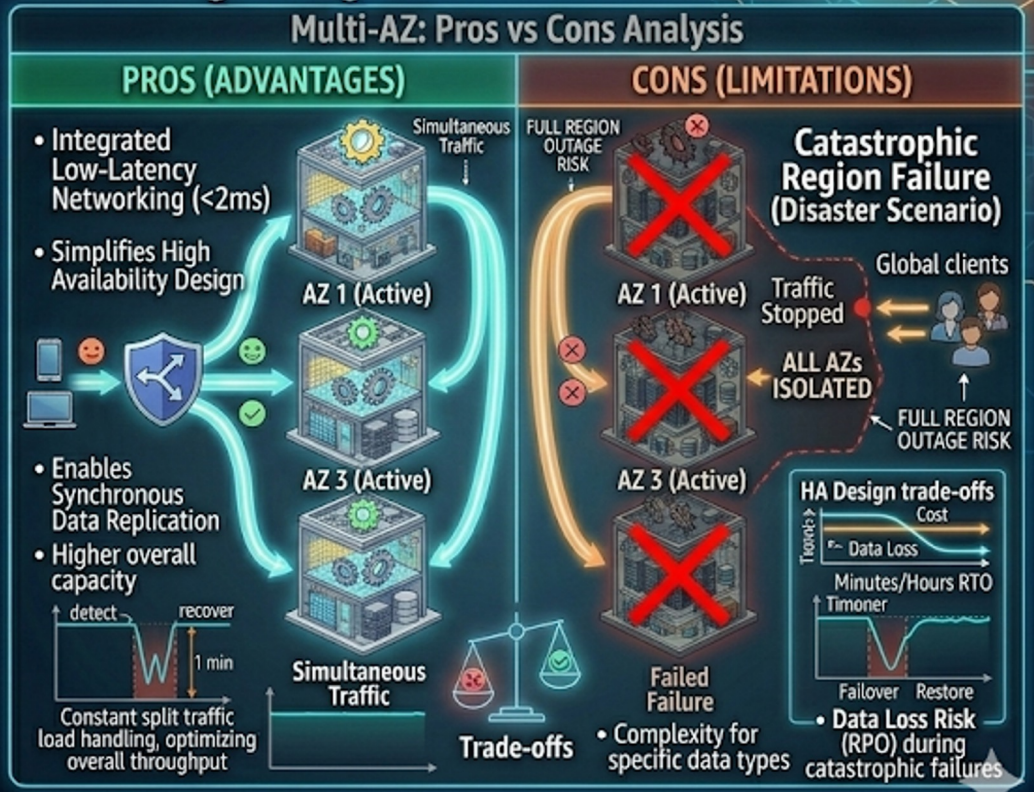
### Integrated Local High Availability

- low latency
- easy (simple implementation)

### Primary Drawback

- region outage risk
- Higher costs & complexity for certain configurations
- Designed to prevent single data center failure from impacting service availability.

**Final thought:** Multi-AZ architecture is the foundation for local high availability, but does not protect against catastrophic regional events.



# MULTI-REGION: Analysis and Trade-offs

## Understanding Global Infrastructure Resilience

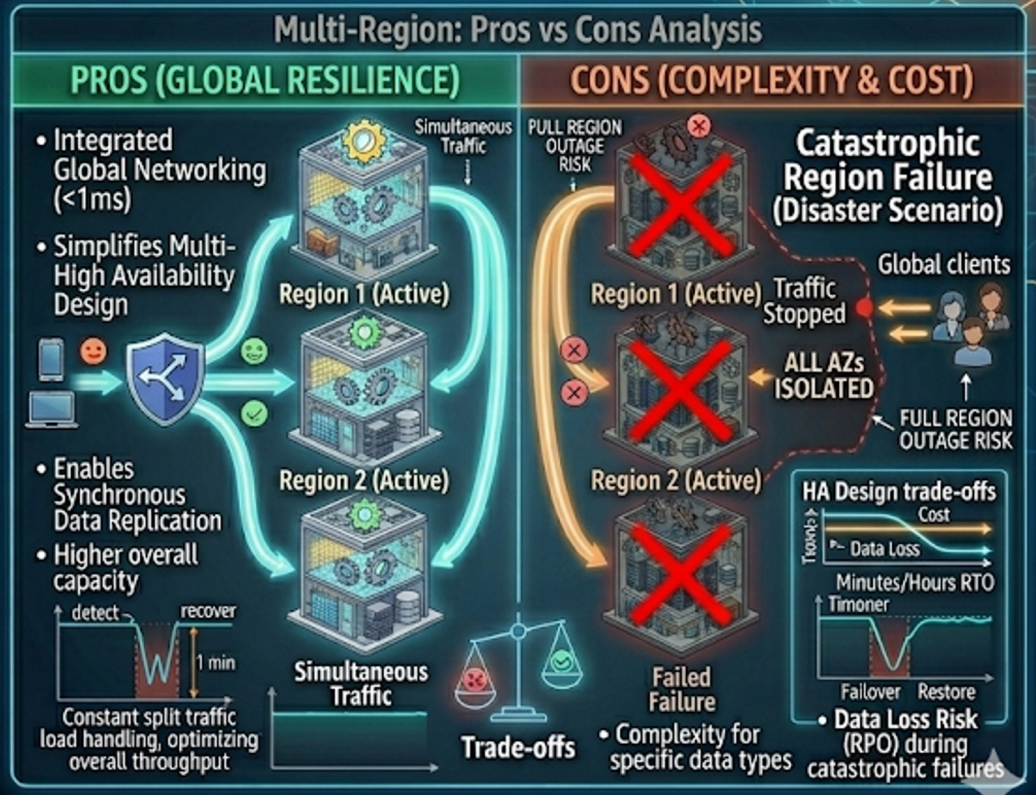
### Global Resilience & Business Continuity

- disaster resilience (protects from catastrophic regional events)
- reduced latency for global users

### Primary Drawbacks

- complexity (global networking, data synchronization)
- cost (duplicated resources, inter-region data transfer)

**Final thought:** Multi-region architecture is critical for protecting against global disasters and catering to worldwide audiences, but requires careful consideration of its significant cost and operational complexity.

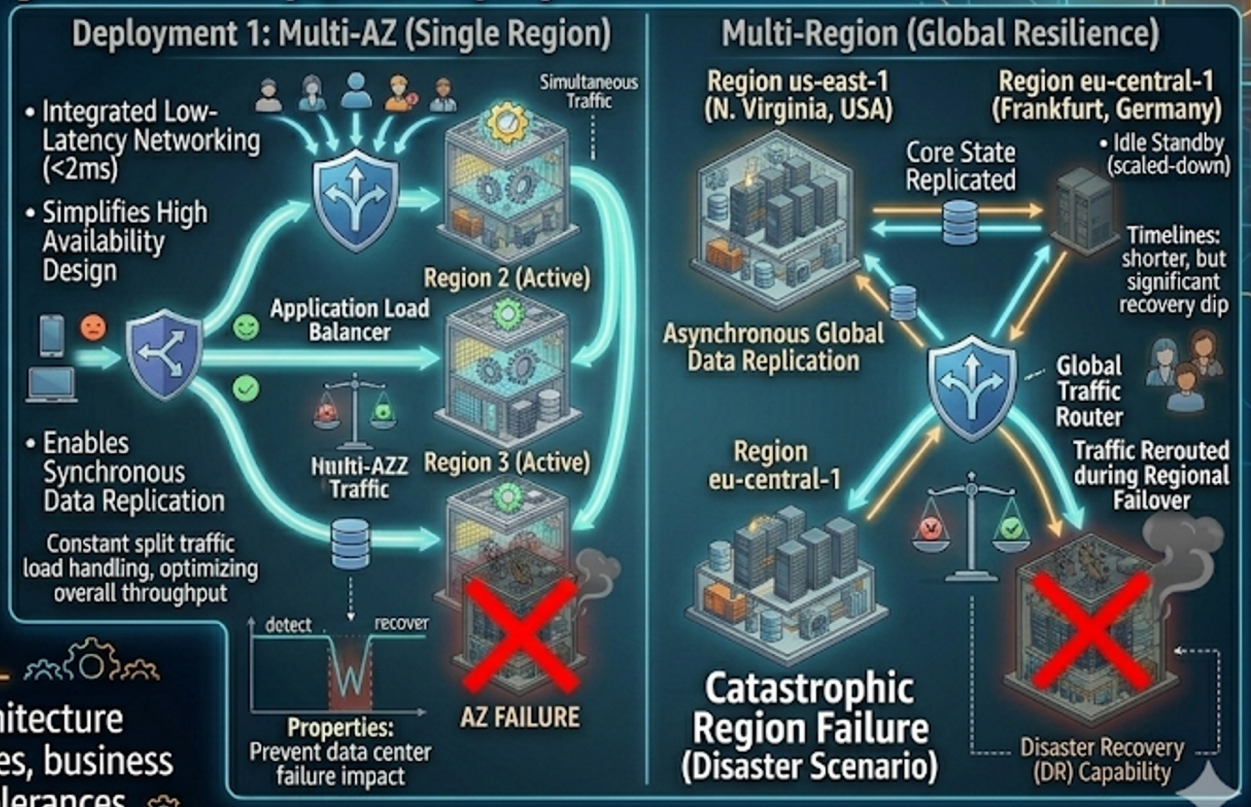


# Architecture Comparison

## Multi-AZ vs Multi-Region High Availability (HA) Deployments

### Selecting the Optimal Deployment Strategy

- Choice impacts RTO, RPO, cost, and complexity.
- Multi-AZ prevents data center failure impact.
- Multi-Region provides disaster recovery (DR) capability.
- Traffic routing ensures minimal downtime during failover.




**Final thought:** Select your HA architecture based on defined RTO/RPO objectives, business continuity requirements, and cost tolerances.

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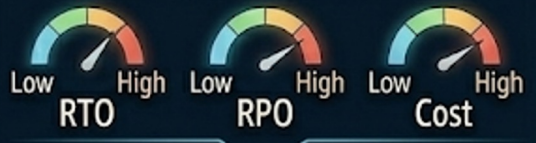
# AWS HIGH AVAILABILITY (HA) DECISION GUIDE:
## When to Use Multi-AZ vs. Multi-Region

**Factors:** Choice impacts RTO, RPO, cost, and complexity.
**Value:** Multi-AZ prevents Data Center (DC) failure impact.
Multi-Region provides Disaster Recovery (DR) capability.

### STEP 1: Define Goals




### STEP 2: Trade-offs Matrix

| FACTOR | Multi-AZ (Single Region) | Multi-Region (Global Resilience) |
|---|---|---|
| Latency | Integrated <2ms | Dependent on Distance |
| Cost | Moderate to Low | High |
| Complexity | Low | Very High |
| Sync | Synchronous | Asynchronous |
| RTO/RPO (Time/Loss) | 1 min 1 min RPO | Region RPO |

### STEP 3: Recommendations

**Choose Multi-AZ if:**
- Primary goal is preventing DC failure
- Need ultra-low latency within application stack
- Synchronous data replication requirements
- Cost/complexity tolerance is low
- Single-region workload scope

**Choose Multi-Region if:**
- Primary goal is global disaster recovery
- Globally distributed users with latency needs
- Required multi-region state consistency
- High tolerance for cost and operational complexity
- Workload demands global reach

**Final thought:** Choose based on RTO/RPO objectives, business continuity, and cost. Traffic routing ensures minimal downtime during failover.
Active/Active flow is Depicted in the original comparison.

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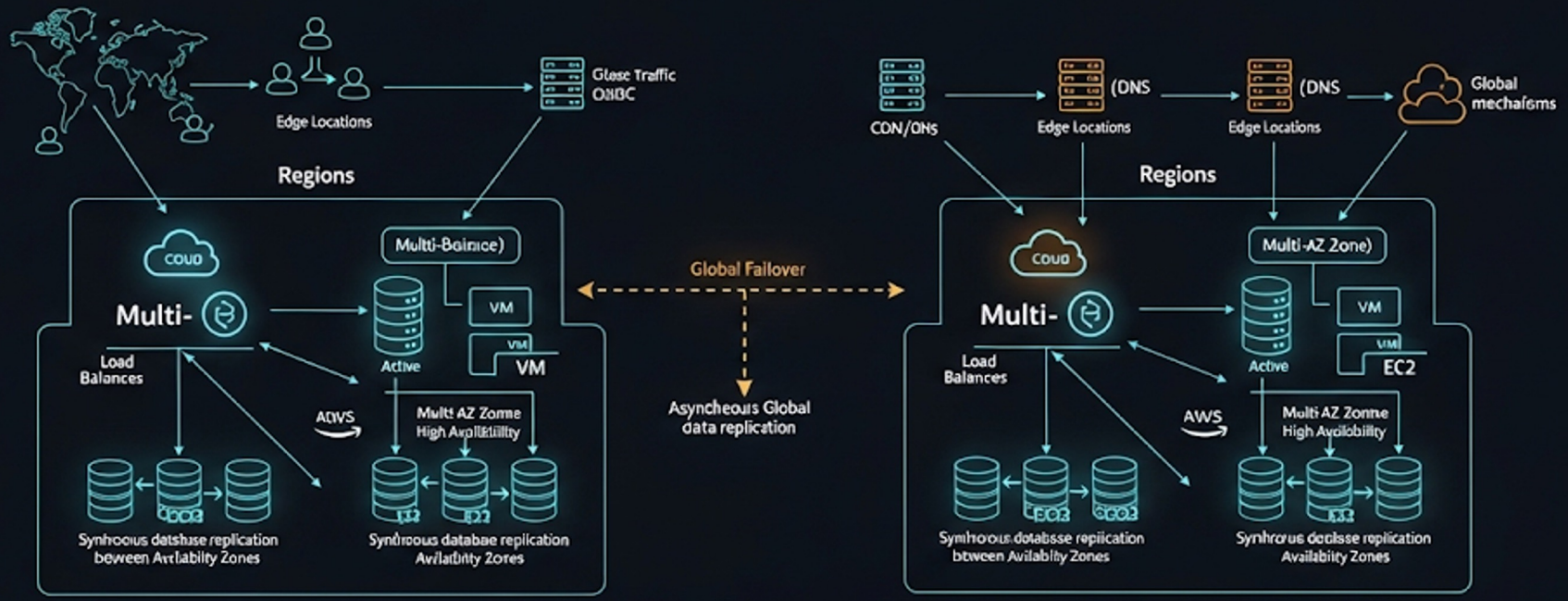
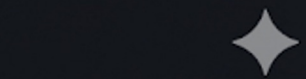
# GLOBAL ARCHITECTURE OVERVIEW



# Key Takeaways

- Regions provide geographic isolation
- AZs provide high availability
- Edge locations reduce latency



# Checklist: What Students Should Understand

- Region vs AZ
- Global vs Regional Services
- HA Patterns
- Multi-AZ vs Multi-Region